

Figuras candidatas v3 — Institutional Map e alternativas

2026-04-27

Objetivo

Quatro novas candidatas para a Figura 4 do v5, todas mostrando diretamente a Proposition (classificacao institucional). Parametros do Example: $N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$.

Candidata 5: Institutional map em (p, c)

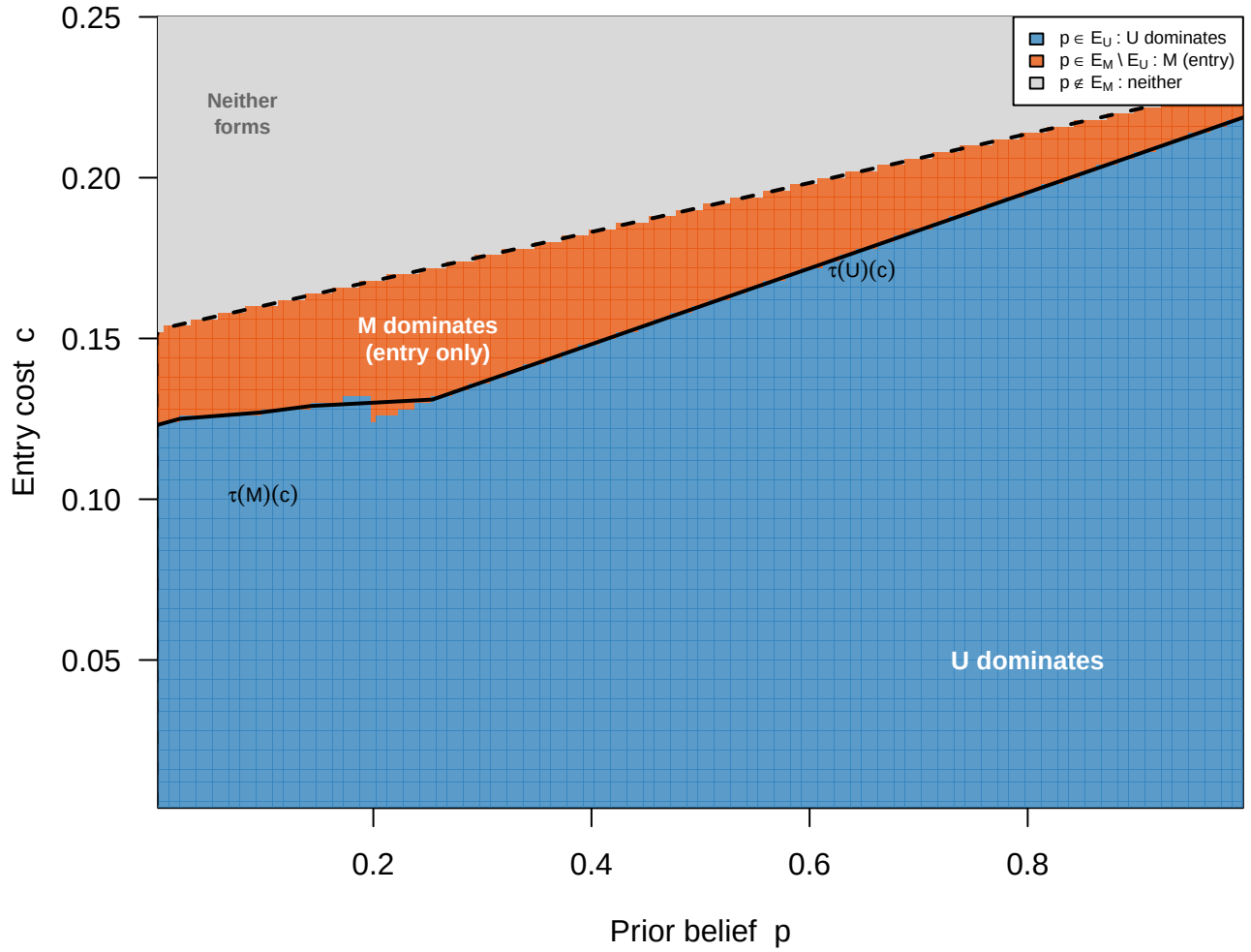


Figure 1: Institutional classification in the (p, c) plane ($N = 5, r = 1.5, \alpha = 0.3, \beta = 0.9$). Blue: unanimity dominates ($p \in E_U$). Orange: majority dominates through wider entry ($p \in E_M \setminus E_U$). Gray: neither rule sustains entry ($p \notin E_M$). The boundary between the blue and orange regions traces the unanimity entry threshold $\tau(U)$; the boundary between the orange and gray regions traces $\tau(M)$. Since $E_U \subseteq E_M$, the unanimity-dominance region is always nested inside the majority-entry region. Conditional on entry, unanimity strictly dominates at every belief (Theorem 1).

O que mostra: a Proposition como geometria no plano (p, c) . A regioazul (unanimidade) esta sempre aninhada dentro da laranja+azul (maioria). A “fatia” laranja entre as fronteiras e o scope advantage da maioria. Nao fixa c arbitrariamente — mostra todos os c de uma vez.

Candidata 6: Two-panel institutional mechanism

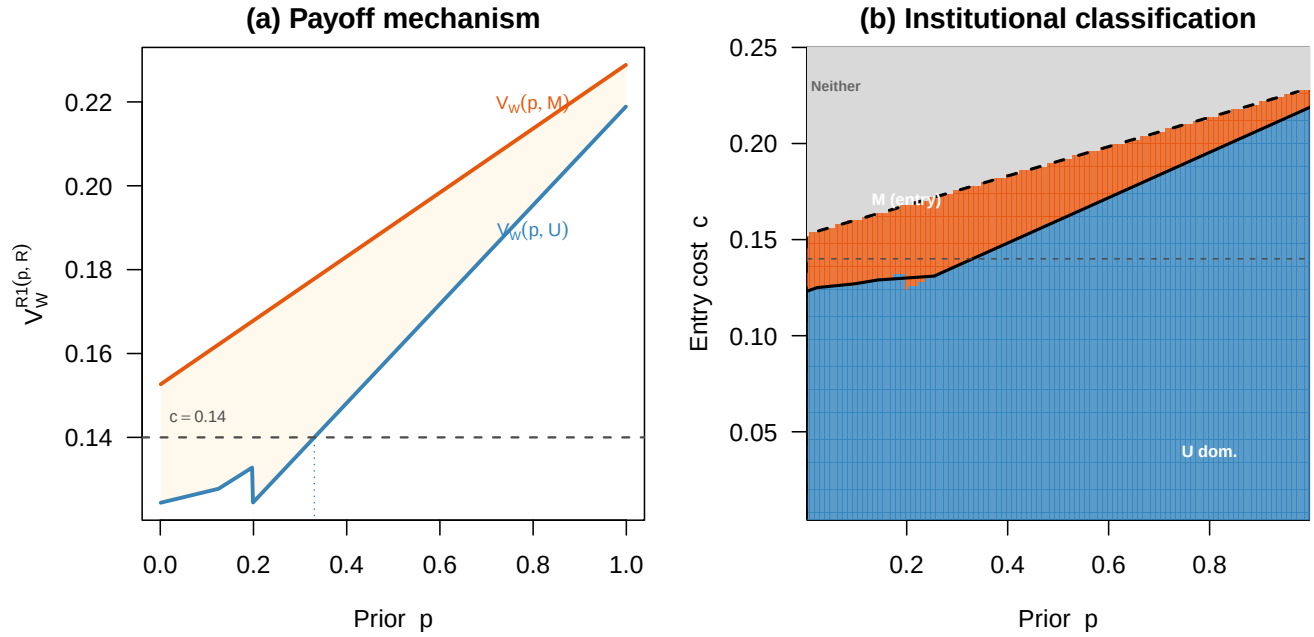


Figure 2: Panel (a): Weak-state payoffs under majority ($V_W^{R1}(p, M)$, orange) and unanimity ($V_W^{R1}(p, U)$, blue). The gap between the curves reflects the screening rent captured by the hegemon under unanimity (Theorem 1). A single entry cost $c = 0.14$ (dashed) defines the entry thresholds $\tau(M)$ and $\tau(U)$. Panel (b): Institutional classification in the (p, c) plane. Blue: unanimity dominates ($p \in E_U$). Orange: majority dominates through wider entry ($p \in E_M \setminus E_U$). Gray: neither rule sustains entry. Parameters: $N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$.

O que mostra: Panel A explica *por que* $E_U \subseteq E_M$ (payoff curves + uma unica linha c). Panel B mostra a classificacao completa no plano (p, c) . A linha $c = 0.14$ conecta os dois paineis: no Panel A e a horizontal que define os thresholds; no Panel B e o corte que gera uma “barra” equivalente a Candidata 3.

Candidata 7: Scope advantage (areas empilhadas)

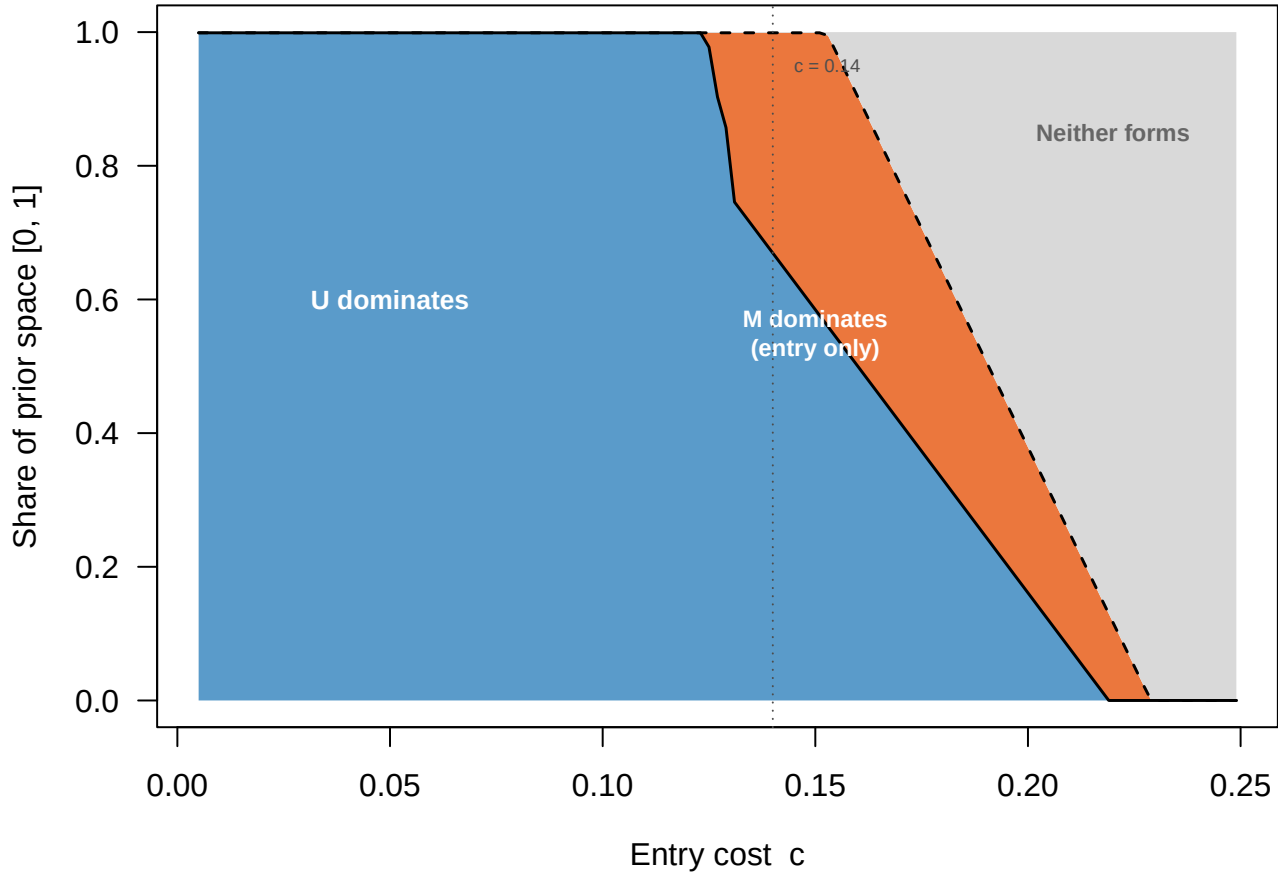


Figure 3: Share of the prior space in each institutional regime as a function of entry cost c ($N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$). Blue: fraction of priors where unanimity dominates ($|E_U|$). Orange: fraction where majority dominates through wider entry ($|E_M \setminus E_U|$). Gray: fraction where neither rule sustains entry ($|[0, 1] \setminus E_M|$). As c increases, majority's scope advantage grows, but this advantage is entirely through the entry channel: conditional on participation, unanimity strictly dominates at every belief (Theorem 1).

O que mostra: comparative statics em c mostrando diretamente o tamanho de cada regime. A area azul (U dominates) encolhe com c ; a laranja (M entry advantage) cresce. A soma e sempre 1. Preserva a Proposition e permite comparative statics. Nota: assume E_U e E_M sao intervalos (conectados) para calcular $|E_U| = 1 - \tau(U)$.

Candidata 8: Scope advantage em α (areas empilhadas)

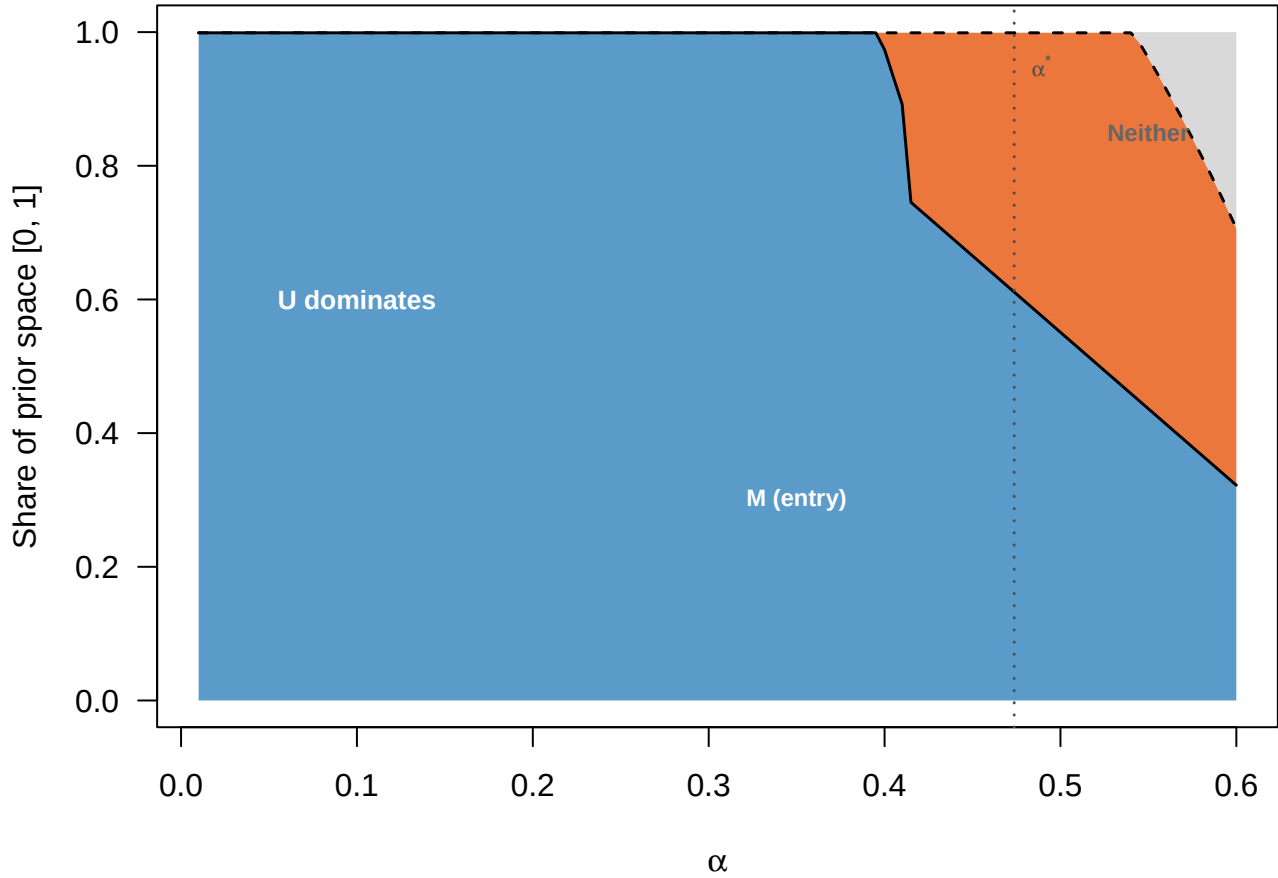


Figure 4: Share of the prior space in each institutional regime as a function of the hegemon's outside-option strength α ($N = 5$, $r = 1.5$, $\beta = 0.9$, $c = 0.1$). Blue: fraction of priors where unanimity dominates ($|E_U|$). Orange: fraction where majority dominates through wider entry ($|E_M \setminus E_U|$). Gray: fraction where neither rule sustains entry. Dashed vertical line: α^* , below which Theorem 1 guarantees conditional dominance at every belief. As α increases, both entry sets shrink, but unanimity shrinks faster because stronger outside options raise the cost of buying unanimous agreement.

O que mostra: o mesmo conceito de areas empilhadas, mas variando α em vez de c . Mostra que α^* nao e uma fronteira dramatica: a area azul ja vinha encolhendo gradualmente, e continua a encolher apos α^* . O scope advantage da maioria cresce com α porque outside options mais fortes tornam a unanimidade mais cara.

Resumo comparativo

Candidata	Tipo	Mostra	Comparative statics	Risco
5: Map (p, c)	Heatmap 2D	Classificacao completa	Em c (contínuo)	Nenhum
6: Two-panel	V_W + Map	Mecanismo + classificacao	Em c (contínuo)	Mais complexo
7: Scope vs c	Areas empilhadas	Tamanho dos regimes	Em c	Assume E_U conectado
8: Scope vs α	Areas empilhadas	Tamanho dos regimes	Em α	Assume E_U conectado

Recomendacao: Candidata 5 (Map (p, c)) como Figura 4 principal, ou Candidata 6 (two-panel) se quiser explicar o mecanismo na mesma figura.